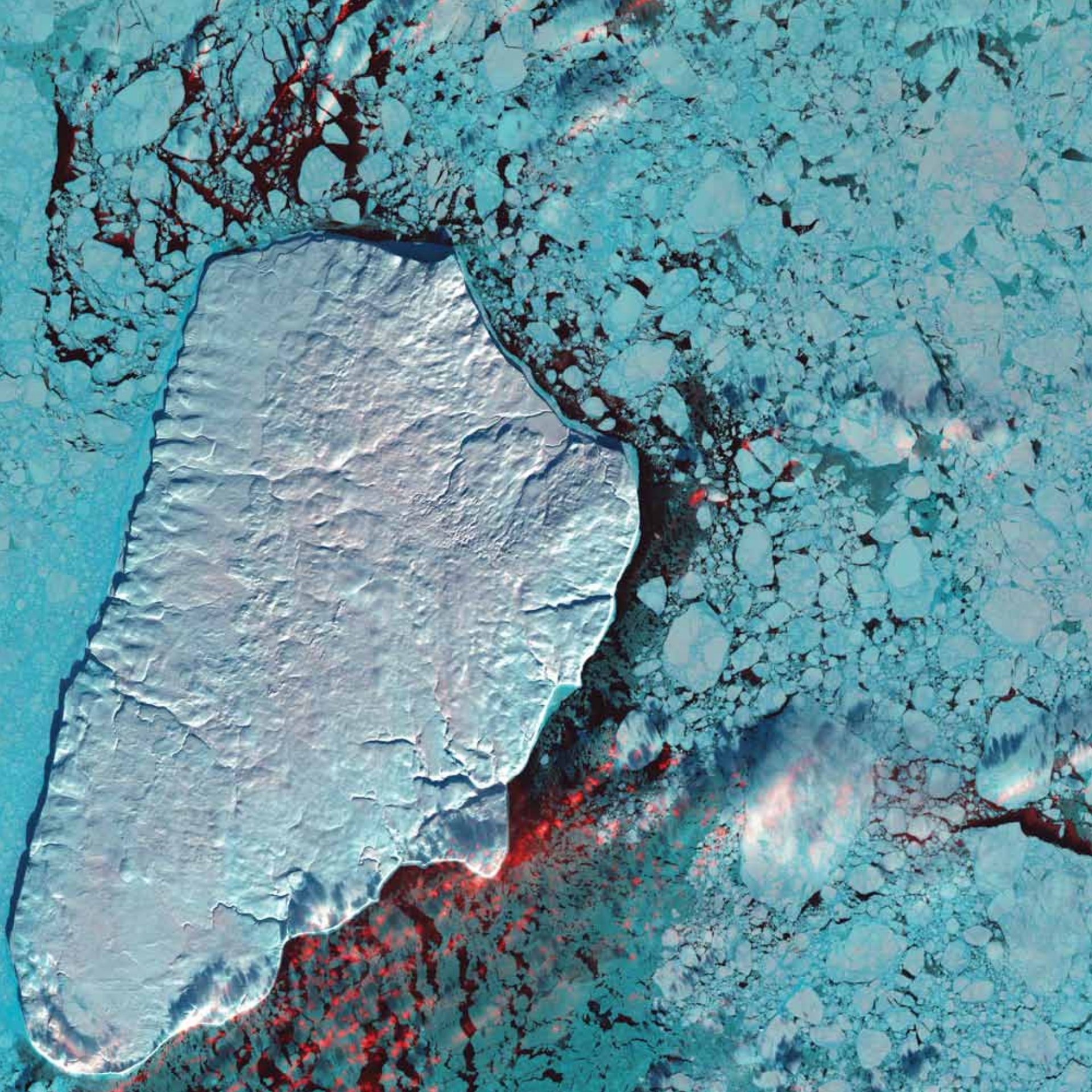


Akpatok Island
Canada

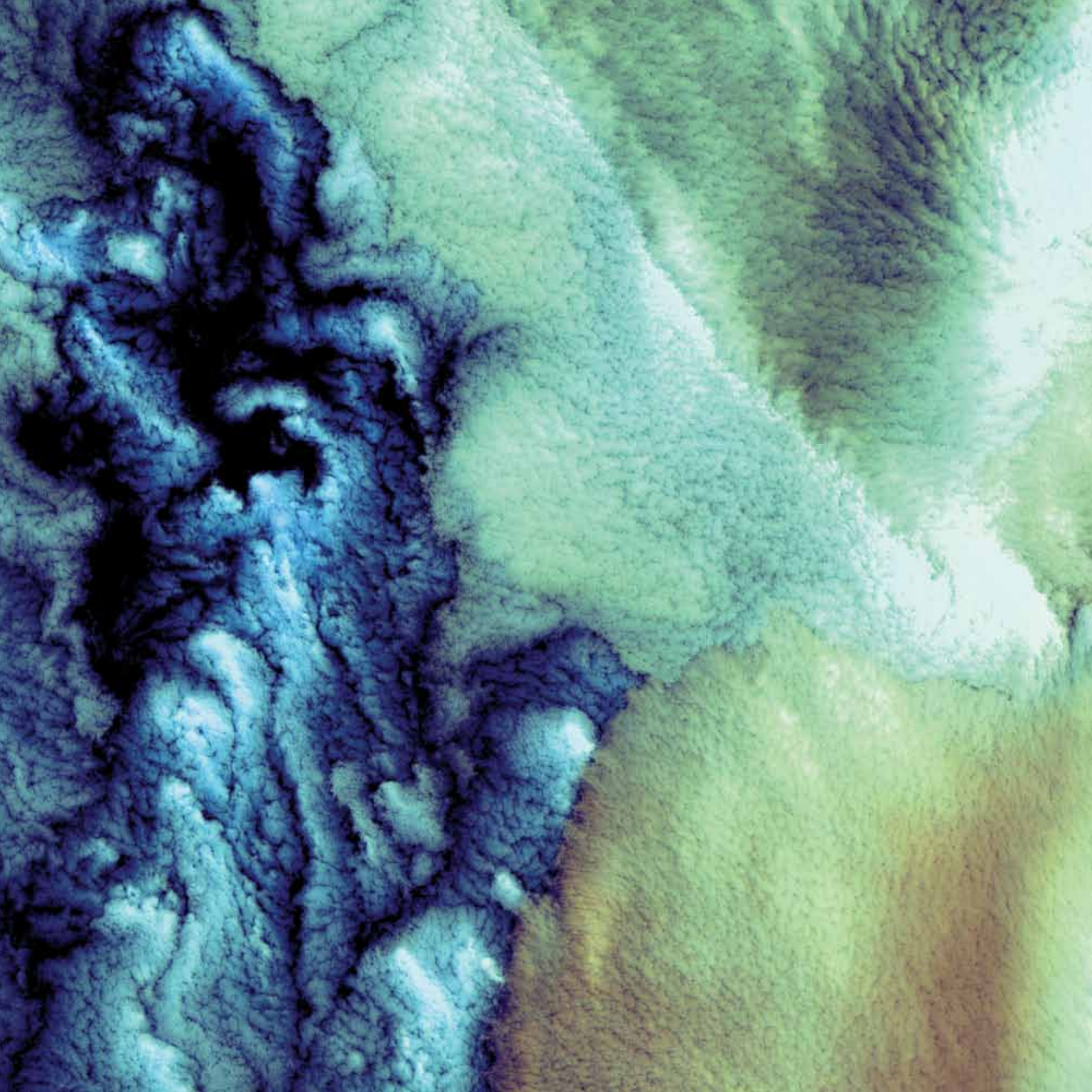
Akpatok Island rises sharply out of the frigid water of Ungava Bay in northern Quebec, Canada. Composed primarily of limestone, the island is a flat, treeless plateau 23 kilometers wide, 45 kilometers long, and about 150 to 250 meters high. This 2001 Landsat 7 image shows Akpatok Island completely covered in snow and ice. Small, dark patches of open water appear between pieces of pale blue-green sea ice, and a few scattered clouds are shown in red. The surrounding sea and ice are home to polar bears, walruses, and whales. A traditional hunting ground for native Inuit people, Akpatok is almost inaccessible except by air. The island is an important sanctuary for seabirds that make their nests in the steep cliffs that circle the island.





Aleutian Clouds
Bering Sea

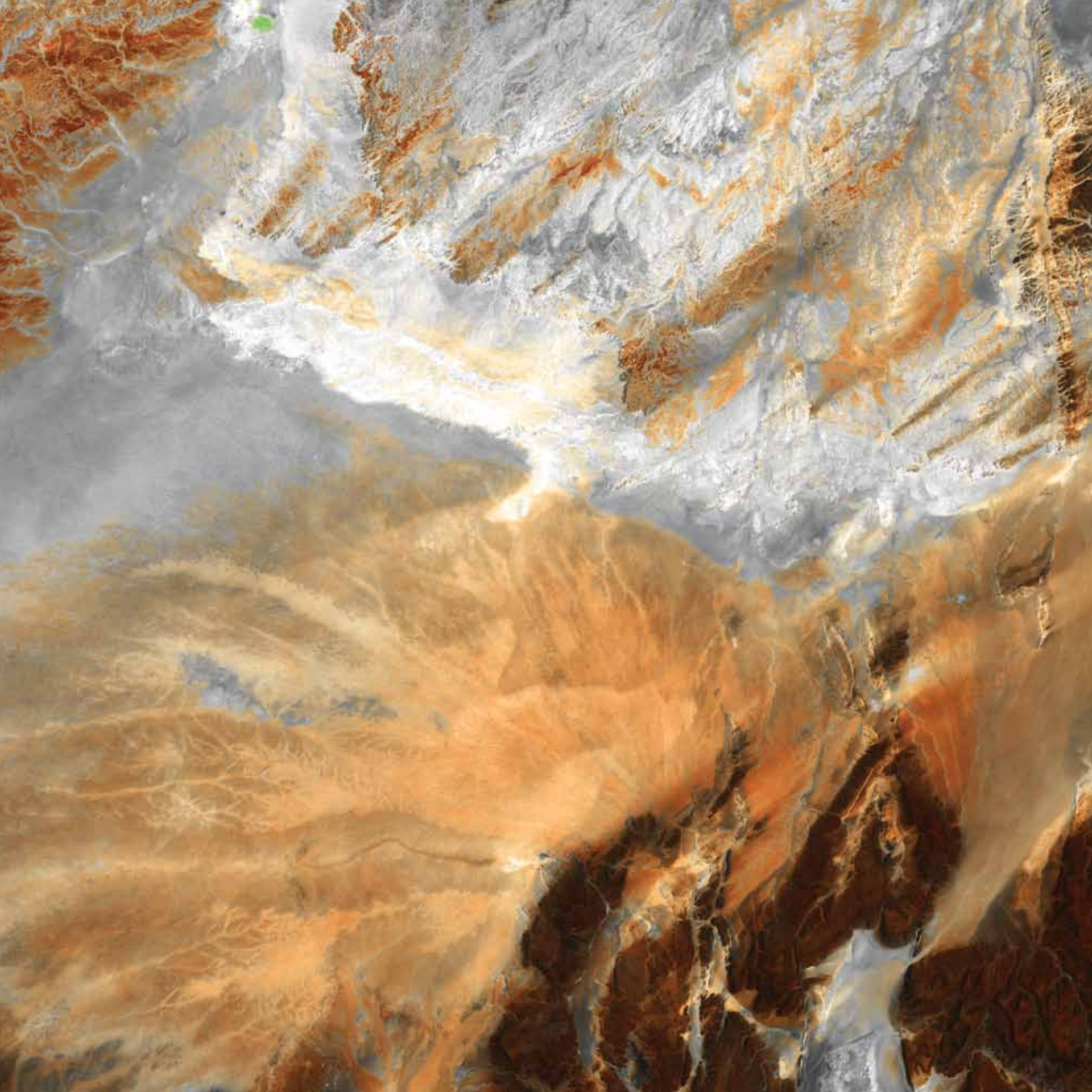
Clouds hover over the waters off the western Aleutian Islands, where fog, heavy rains, and high winds are common. While the clouds in this 2000 Landsat 7 image are structured differently, all the clouds shown are low, marine stratocumulus clouds, which often produce drizzle. The color variations are probably due to differences in the temperature and in the size of the water droplets that make up the clouds. The Aleutian Islands are part of the Pacific Ring of Fire. The archipelago curves out 1,800 kilometers from southwestern Alaska towards Russia's Kamchatka Peninsula.





Algerian Desert
Algeria

Lying amid the Great Eastern Erg, the Great Western Erg, and the Atlas Mountains in Northern Africa, the Sahara Desert in central Algeria is dotted by fragmented mountains (in brown, lower right) where barren, windswept ridges overlook arid plains. In this Landsat 5 image from 2009, a system of dry streambeds crisscrosses the rocky landscape awaiting the rare, intense rains that often cause flash floods.





Alluvial Fan
China

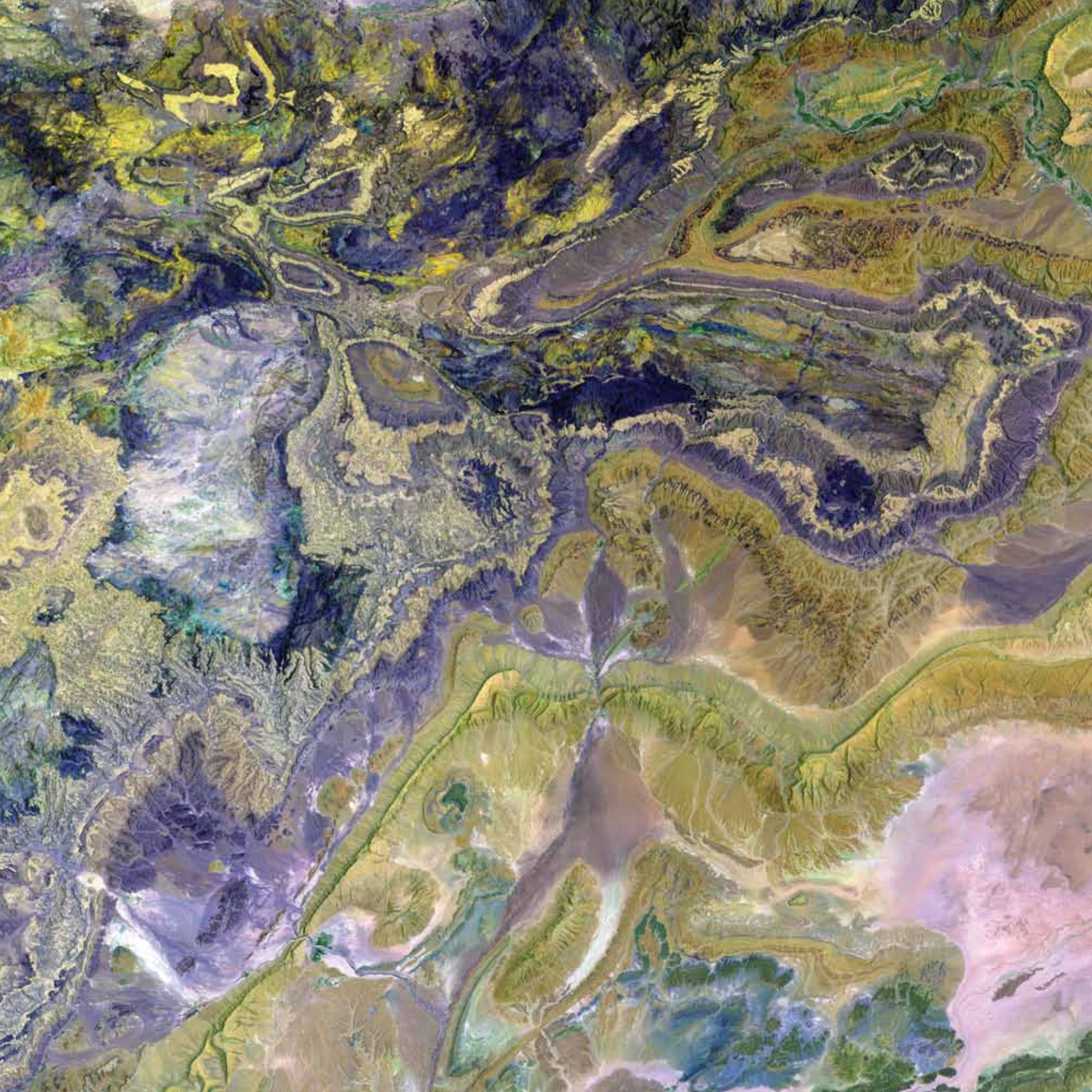
A vast alluvial fan unfolds across the desolate landscape between the Kunlun and Altun mountain ranges that form the southern border of the Taklimakan Desert in China's Xinjiang Province. The fan is about 60 kilometers long and 55 kilometers wide at its broadest point. The left side is the active part of the fan. Water flowing down from the mountains in the many small streams appears blue in this 2002 image from the Terra satellite. Vegetation appears red and can be seen in the upper left corner of the image. Farmers take advantage of water at the foot of the fan to irrigate small fields. The "lumpy" terrain at the top of the image is composed of sand dunes at the edge of the Taklimakan, one of the largest sandy deserts on Earth. Shifting sand dunes, some reaching as high as 200 meters, cover more than 80 percent of the desert floor.





Anti-Atlas Mountains
Morocco

A part of the Atlas Mountains in northwest Africa, the Anti-Atlas range runs for several hundred kilometers. The range extends from the Atlantic Ocean in southwest Morocco toward the northeast, where it meets the High Atlas range closer to the Mediterranean Coast. The Anti-Atlas mountains formed as a result of continental collisions between 65 and 250 million years ago, which destroyed the then Tethys Ocean. The limestone, sandstone, claystone, and gypsum layers that formed the ocean bed were folded and crumpled to create the mountains. This Landsat 7 image from 2001 highlights some of the different rock types and illustrates the complex folding.





Anyuyskiy Volcano
Russia

Anyuyskiy Volcano lies north of the Kamchatka Peninsula in eastern Russia. Now dormant, the volcano was once active enough to send a massive lahar (an avalanche of volcanic ash and rock mixed with water) 50 kilometers down the west side of the volcano summit. The dried, hardened remains of the lahar persist today as a streak of barren rock on a landscape that is otherwise richly vegetated. In this Landsat 7 image from 2001, vegetation appears green, bare rock and ice appear bright red, and water appears navy blue. The Anyuyskiy lahar extends from the volcano's north slope, turns sharply westward, and flows toward the west-southwest. Lakes occur along the margins of the lahar, and some small lakes appear on the lahar's surface. Little vegetation has encroached on the ancient river of rock. Remote and largely inaccessible, the region is a rugged collection of towering volcanic peaks, steep valleys, and snow-fed rivers and streams.